

Laboratory Exercise: Phylogenetic Analysis Using NCBI and Galaxy

Group 1-Felidae Members:

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Introduction

Reconstructing evolutionary relationships among taxa is a fundamental objective of systematic biology, providing crucial insights into lineage divergence, adaptive radiation, and biodiversity. Modern molecular phylogenetics relies on nucleotide sequence variation in targeted genetic markers to infer common ancestry across evolutionary time scales (Yu et al., 2025). Among these markers, the mitochondrial cytochrome c oxidase subunit I (COI/COX1) gene is widely utilized because of its maternal inheritance, low rate of recombination, and optimal balance between conserved and variable regions (Lee & Vogler, 2026). These characteristics make COI a valuable molecular marker for species identification through DNA barcoding and for investigating evolutionary relationships among closely related taxa.

In this laboratory exercise, phylogenetic relationships within the family Felidae were investigated by analysing representative species from two major evolutionary lineages: the small cats of the genus *Felis* (*Felis catus*, *Felis chaus*, and *Felis silvestris*) and the big cats of the genus *Panthera* (*Panthera pardus*, *Panthera tigris*, and *Panthera uncia*). To root the phylogenetic tree and provide an evolutionary reference for comparison, the red fox (*Vulpes vulpes*; Family Canidae) was included as an outgroup.

Authentic mitochondrial COX1 nucleotide sequences were retrieved from the National Center for Biotechnology Information (NCBI) Nucleotide database. Individual FASTA sequences were compiled into a multi-FASTA dataset and aligned using MAFFT to identify homologous nucleotide positions and generate a consensus sequence. A maximum-likelihood phylogenetic tree was then reconstructed using FastTree within the Galaxy bioinformatics platform to infer the evolutionary relationships among the selected taxa.

Accordingly, this study addresses the biological question: To what extent can mitochondrial COX1 nucleotide sequences accurately infer the phylogenetic relationships among selected species of the family Felidae? By evaluating the resulting phylogenetic tree, the study also examines the strengths and limitations of using a single mitochondrial marker for evolutionary inference.

Reconstructing Evolutionary Relationships Using Mitochondrial COX1

Objective: Retrieve authentic mitochondrial cytochrome c oxidase subunit I (COI/COX1) sequences from NCBI, align them in Galaxy, examine a consensus sequence, construct a phylogenetic tree, and compare the resulting relationships with published studies.

Learning Outcomes

- Retrieve homologous COI sequences and accession numbers from NCBI Nucleotide.
- Create and manage a free usegalaxy.org account and History.
- Perform multiple sequence alignment using MUSCLE.
- Generate and interpret a consensus sequence.
- Construct and interpret a phylogenetic tree from aligned nucleotide sequences.
- Compare a COI phylogeny with peer-reviewed published accounts.

Required Sequence Information Table

Scientific name	Common name	NCBI accession	Length (bp)	Focal taxon / Outgroup
<i>Felis catus</i>	Domestic cat	FJ185310	732	Ingroup
<i>Felis chaus</i>	Jungle cat (swamp cat)	OR150429	665	Ingroup
<i>Felis silvestris</i>	Wildcat (European wildcat)	OR178528.1	658	Ingroup
<i>Panthera pardus</i>	Leopard	PP527113.1	643	Ingroup
<i>Panthera tigris</i>	Tiger	KP992945.1	516	Ingroup
<i>Panthera uncia</i>	Snow leopard	ON377296.1	662	Ingroup
<i>Vulpes vulpes</i>	Red fox	PP527100	643	Outgroup

Table 1. Taxonomic, accession, and sequence data for mitochondrial COX1 genes analyzed.

Sequence Retrieval from NCBI

Mitochondrial cytochrome c oxidase subunit I (COX1) nucleotide sequences were retrieved from the National Center for Biotechnology Information (NCBI) Nucleotide database. Each target species was searched individually using the organism name together with the keywords "COX1" and "mitochondrion." Sequence records were verified to ensure they corresponded to the mitochondrial COX1 gene, represented homologous regions, and had comparable sequence lengths. The accession number, sequence length, and record title of each selected sequence were recorded. All sequences were downloaded in FASTA format without modifying the nucleotide sequences.

Data Preparation in Galaxy

The downloaded FASTA sequences were uploaded to the Galaxy platform (<https://usegalaxy.org>) using a dedicated analysis history. Individual FASTA files were combined into a single multi-FASTA dataset containing one sequence for each species.

FASTA headers were renamed with short, unique species labels to improve readability in the final phylogenetic tree while preserving the original sequence data.

Multiple Sequence Alignment

Because the MUSCLE alignment tool was unavailable on the Galaxy server, multiple sequence alignment was performed using MAFFT, an alternative alignment program integrated into Galaxy. The combined multi-FASTA dataset was used as the input, and the default alignment parameters were applied. The resulting alignment was inspected to confirm that homologous nucleotide positions were properly aligned and to identify any sequences with poor alignment, excessive missing data, incorrect orientation, or evidence of non-homologous regions. The finalized aligned dataset was used for all subsequent analyses.

Consensus Sequence Generation

A consensus sequence was generated from the MAFFT alignment using the consensus sequence tool available in Galaxy with the default majority-rule settings. The resulting consensus sequence was examined to determine its length and to identify conserved and variable nucleotide sites across the aligned sequences. The consensus sequence served only as a summary of the alignment and was not used for phylogenetic tree construction.

Phylogenetic Tree Construction

Phylogenetic relationships among the selected taxa were inferred using the FastTree tool in Galaxy. The aligned COX1 nucleotide sequences produced by MAFFT were used as the input, with nucleotide sequence analysis selected where applicable. FastTree generated a phylogenetic tree in Newick format, which was visualized using a compatible tree viewer. Species labels were checked for clarity, and the designated outgroup was used to aid in rooting and interpreting the evolutionary relationships among the sampled taxa.

MAFFT alignment

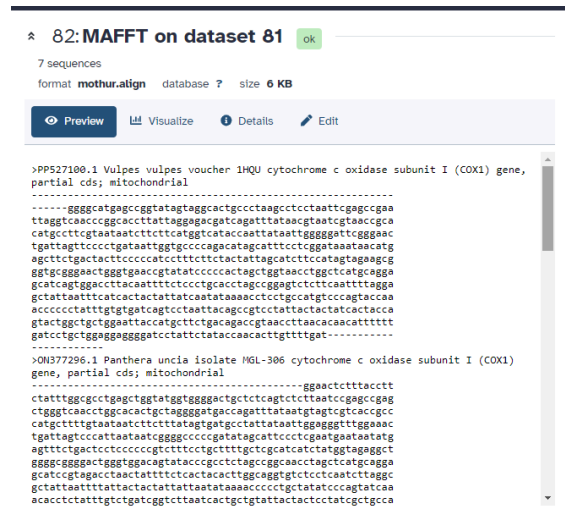


Figure 1. Multiple sequence alignment of mitochondrial COX1 nucleotide sequences using MAFFT as an alternative to MUSCLE in the Galaxy platform.

Consensus sequence



Figure 2. Consensus sequence mitochondrial COX1 nucleotide sequences using Galaxy platform.

Phylogenetic tree

The phylogenetic tree groups the species into two major clades corresponding to the genera *Felis* and *Panthera*, with *Vulpes vulpes* (red fox) serving as the outgroup. The outgroup branches off first, indicating that it is more distantly related to the cat species and providing a reference point for interpreting the evolutionary relationships within the family Felidae.

Within the *Felis* clade, *Felis silvestris* and *Felis chaus* are sister taxa, indicating that they share the most recent common ancestor with each other. This pair forms a larger clade with *Felis catus*, suggesting that the domestic cat is closely related to these two wild cat species.

Within the *Panthera* clade, *Panthera uncia* (snow leopard) and *Panthera pardus* (leopard) are sister taxa, sharing the most recent common ancestor. *Panthera tigris* (tiger) branches earlier and is therefore more distantly related to the snow leopard–leopard pair than those two species are to each other.

The branch lengths represent the estimated amount of evolutionary change or genetic divergence based on the COX1 nucleotide sequences. Shorter branches indicate fewer nucleotide substitutions and greater genetic similarity, whereas longer branches indicate greater evolutionary divergence. The scale bar (0.03) represents approximately 0.03 substitutions per nucleotide site.

The support values shown at the internal nodes indicate the confidence of the inferred relationships. Most nodes have high support values (0.859–1.000), suggesting that the major clades and sister-taxon relationships are well supported by the COX1 sequence data. Overall, the tree is consistent with the expected evolutionary relationships among members of the family Felidae, with the fox appropriately placed as the outgroup outside the cat lineage.

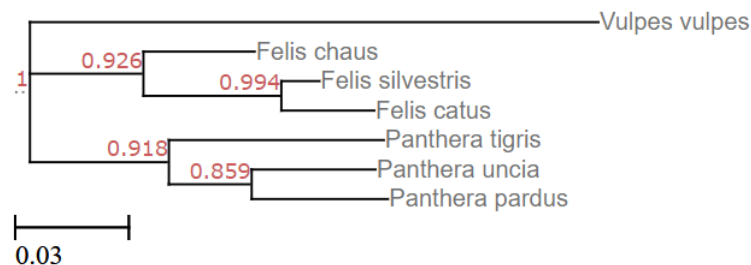


Figure 3. Phylogenetic tree in Newick format of selected Felidae species based on mitochondrial COX1 sequences generated using FastTree.

To evaluate the accuracy of the phylogenetic relationships inferred from the mitochondrial cytochrome c oxidase subunit I (COX1) gene, the resulting phylogenetic tree was compared with the phylogenomic study of Li et al. (2016), which reconstructed the evolutionary history of extant Felidae using genome-wide nuclear DNA, complete mitochondrial genomes, and sex chromosome markers. Unlike the present study, which relied solely on a single mitochondrial marker (COX1), Li et al. (2016) employed a comprehensive genomic dataset to resolve evolutionary relationships and investigate the effects of historical hybridization among felid lineages.

The COX1 phylogenetic tree recovered two well-supported monophyletic clades corresponding to the genera *Felis* and *Panthera*. The *Felis* clade comprised *Felis chaus*, *Felis silvestris*, and *Felis catus*, whereas the *Panthera* clade consisted of *Panthera tigris*, *Panthera uncia*, and *Panthera pardus*. This topology is consistent with the phylogeny presented by Li et al. (2016), which recognized both genera as distinct evolutionary lineages within the family Felidae. The placement of *Vulpes vulpes* outside the Felidae clade further supports the use of the red fox as an appropriate outgroup for rooting the phylogenetic tree.

Within the *Felis* clade, the present study recovered *Felis silvestris* and *Felis catus* as sister taxa, supported by a local support value of 0.994, while *Felis chaus* diverged earlier within the clade. This relationship agrees with Li et al. (2016), who identified domestic cats and wildcats as members of the Domestic Cat lineage and demonstrated their close evolutionary affinity using genome-scale molecular data. The concordance between the COX1 phylogeny and the genome-wide analysis indicates that the mitochondrial COX1 gene effectively resolves evolutionary relationships among closely related members of the genus *Felis*.

Similarly, the *Panthera* clade was recovered as a distinct monophyletic lineage. In the present analysis, *Panthera uncia* and *Panthera pardus* formed a sister group with a support value of 0.859, whereas *Panthera tigris* diverged earlier within the clade. This overall grouping agrees with Li et al. (2016), which consistently recognized *Panthera* as a natural evolutionary lineage. However, Li et al. (2016) reported that the internal relationships among *Panthera* species may vary depending on whether mitochondrial or nuclear genomic datasets are analyzed because ancient hybridization and incomplete lineage sorting have influenced the evolutionary history of the genus. Consequently, slight differences in the branching order among *Panthera* species between the two studies are expected.

The observed similarities indicate that the mitochondrial COX1 gene retains sufficient phylogenetic signal to recover the major evolutionary divisions within Felidae, particularly at the genus level. Nevertheless, several methodological differences explain the minor discrepancies between the present study and the published phylogeny. First, the

current analysis utilized only a single mitochondrial locus inherited maternally, whereas Li et al. (2016) integrated information from genome-wide nuclear markers, complete mitochondrial genomes, and sex chromosomes, providing substantially greater phylogenetic resolution. Second, the published study included 38 extant felid species, while the present analysis examined only six felid species and one canid outgroup, limiting taxonomic representation. Finally, Li et al. (2016) demonstrated that historical hybridization among ancestral felid lineages contributed to discordance between mitochondrial and nuclear phylogenies, a phenomenon that cannot be fully resolved using a single mitochondrial marker.

Overall, the COX1-based phylogenetic tree generated in the present study is highly consistent with the comprehensive phylogenomic framework proposed by Li et al. (2016). Both studies support the separation of *Felis* and *Panthera* into distinct monophyletic lineages and the close relationship between *Felis catus* and *Felis silvestris*. The minor differences observed within the *Panthera* clade most likely reflect the greater resolving power of genome-wide datasets compared with a single mitochondrial gene. Thus, despite its limited genetic scope, the COX1 marker proved effective for reconstructing the major evolutionary relationships among the selected felid species.

Discussion of Agreements, Disagreements, and Limitations of Using COX1 Alone

The COX1 gene is widely accepted as a useful marker for DNA barcoding because it is easy to amplify, shows relatively high variation between species but low variation within species, and is backed by large reference databases. Studies also support its usefulness as a fast and practical first-step tool for species identification in many animal groups.

However, there is also strong disagreement about using COX1 alone for phylogenetic inference. Many authors note that mitochondrial markers can give misleading results when there is introgression, hybridization, incomplete lineage sorting, or the presence of nuclear mitochondrial pseudogenes (NUMTs), because the resulting gene tree may not match the true species tree. In some taxa, COX1 also performs poorly because it does not provide enough resolution or does not amplify consistently, which limits its usefulness outside standard barcoding contexts.

The main limitation of COX1 is that it is a single mitochondrial locus, so it reflects only the maternal lineage and not the full evolutionary history of an organism. Because mitochondrial DNA is non-recombining and maternally inherited, it can be useful for simplifying analyses, but it can also bias interpretation when species histories are complex. COX1 is also subject to functional constraints because it encodes part of a protein involved in respiration, which means its sequence evolution is not always neutral and may be affected by selective pressure.

For these reasons, the most supportable conclusion is that COX1 is excellent for preliminary identification and biodiversity screening, but it should not be used alone for strong phylogenetic conclusions. Many studies recommend combining COX1 with nuclear markers or multilocus datasets to improve accuracy and reduce the risk of misleading results.

Appendices

[Combined raw COX1 FASTA](#)

[Final MAFFT Alignment](#)

[Consensus sequence](#)

[Newick Phylogenetic Tree](#)

Tree image/screenshot

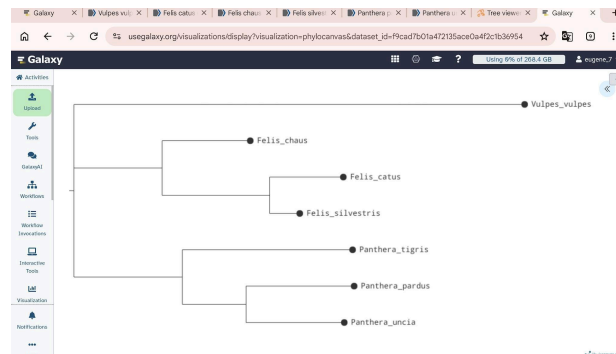
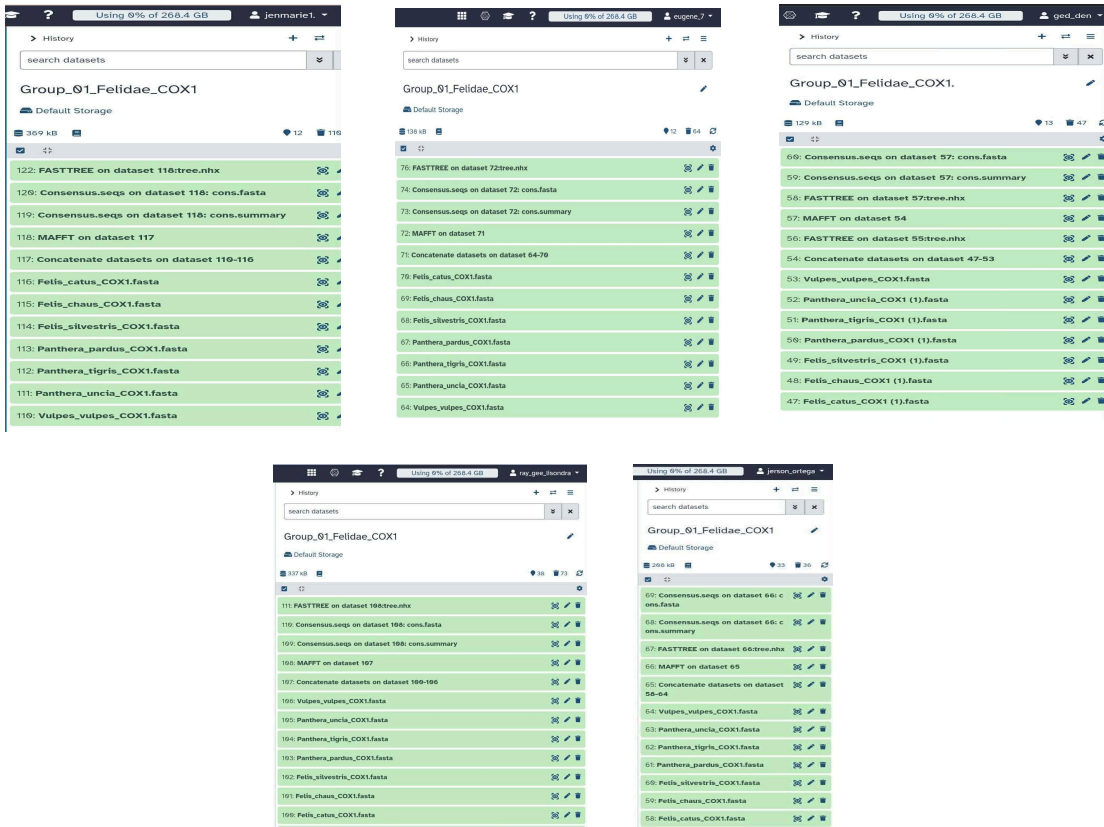


Figure 4. Figure 4. Phylogenetic tree generated using the Galaxy platform based on mitochondrial COX1 gene sequences of selected Felidae species, with Vulpes vulpes as the outgroup.

History and Files uploaded



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